Ritesh Somashekar

AI/ML Systems Analyst & Engineer

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PROFESSIONAL SUMMARY

- **AI/ML Systems Engineer** with **6+ years of experience** building, **testing, and validating** end-to-end **AI/ML, Generative AI, RAG, NLP, and data platforms**, including **3+ years of cloud AI** on **AWS (SageMaker, Bedrock)** and **Azure OpenAI**, with deep expertise in **Python, SQL (Oracle), PySpark, Kafka, LangChain, and LlamaIndex**.
- Proven impact in AI quality, including improving **RAG recall from 61% to 88%**, **reducing LLM hallucination rate by 19%**, sustaining **95%+ production model accuracy** through drift detection, and **cutting reporting errors by 40%** with automated data quality frameworks.
- Specialized in **test strategy for non-deterministic Gen AI systems** (accuracy, relevance, faithfulness, safety, and bias), **BDD (Cucumber/Gherkin), Selenium, Pytest**, API and system testing, and **LLM evaluation frameworks (RAGAS, DeepEval)**; experienced in **Agile** delivery with **Jira and Confluence**, communicating findings to technical and business stakeholders.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL (Oracle, PL/SQL, PostgreSQL, SQL Server), NoSQL, Gherkin
Testing & QA Automation: BDD (Cucumber, Behave, pytest-bdd), Selenium WebDriver, Pytest, Postman, REST API Testing, System, Integration & Regression Testing, Manual & Exploratory Testing, Test Harness & Test Strategy Design
AI/LLM Evaluation & Testing: RAGAS, DeepEval, LLM-as-a-Judge, Golden Datasets, Hallucination Detection, Bias & Fairness Evaluation, Model Drift Monitoring, Prompt Regression Testing, A/B Testing, Embedding & Retrieval Validation
LLM & Generative AI: Large Language Models, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone, pgvector), LangChain, LlamaIndex, LoRA/PEFT Fine-Tuning, Structured Outputs, Embeddings
Machine Learning & Frameworks: ML Model Lifecycle, Deep Learning, NLP, Anomaly Detection, Time Series Forecasting, Feature Engineering, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, XGBoost, LightGBM, Pandas, NumPy
MLOps & Model Deployment: Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, MLflow, Kubeflow, Model Monitoring, Experiment Tracking, Model Serving (FastAPI, Flask), Terraform
Cloud Platforms: AWS (Bedrock, SageMaker, SageMaker Feature Store, S3, Lambda, EC2, Glue, EKS, IAM), Azure (Azure ML, Azure OpenAI, Data Factory, Data Lake), Snowflake, Databricks
Data Engineering: Apache Spark, PySpark, Apache Airflow, Apache Kafka, Apache Flink, dbt, ETL/ELT Pipelines, Data Lakes, Hadoop
Collaboration & Tools: Jira, Confluence, Agile/Scrum, Git, Linux/Unix, Jupyter Notebook, ELK Stack, Power BI, Tableau

PROFESSIONAL EXPERIENCE

Apple | AI/ML Engineer — CA | May 2025 – Present

- Designed an **evaluation harness for non-deterministic LLM outputs** using golden datasets, **RAGAS**, and **LLM-as-a-judge** scoring for accuracy, relevance, faithfulness, and safety, used as a release gate for customer-facing Gen AI features.
- Deployed production LLM applications for customer service automation using **prompt engineering** and **structured outputs**, adding **hallucination detection** and output-validation checks that reduced hallucination rate **19%** and response latency **20%**.
- Built **RAG-based retrieval systems** using **LangChain**, embedding models, and **FAISS** to ground **GPT-4, Claude, and Gemini** outputs in verified sources, validating retrieval relevance and **boosting response accuracy by 16%**.
- Ran **systematic model evaluation and A/B testing** to benchmark LLM quality pre/post-deployment, including **prompt regression and bias evaluation**; logged defects in **Jira** and documented test strategy and findings in **Confluence**.
- Automated **BDD regression suites (pytest-bdd, Gherkin)** and **API tests** for LLM inference endpoints, integrated into **CI/CD** with **MLflow, Docker, and GitHub Actions** to validate every model and prompt release.
- Engineered **real-time data pipelines** with **Apache Spark, Flink, and Kafka** for NLP models, adding automated data validation checks and improving pipeline **throughput by 26%**.
- Fine-tuned transformer models with **PyTorch**, applying **LoRA, PEFT, and knowledge distillation** for intent classification and ticket routing, cutting model size **36%** while holding **87%+ task accuracy** against regression benchmarks.

George Mason University | Graduate Research Assistant (Applied ML) **VA | August 2024 – May 2025**

- Developed a **RAG research assistant** with **LlamaIndex**, prompt engineering, **pgvector**, and **FAISS**, validating embeddings, retrieval, and context assembly to improve **recall@10 from 61% to 88%**.
- Established an **LLM evaluation framework** using **RAGAS, MLflow, Git, and DVC** to measure faithfulness, answer relevance, and context precision, reducing iteration cycles from **3 days to 16 hours**; protocols adopted by **two follow-on LLM studies**.
- Deployed the RAG application with **FastAPI and Docker**, building **automated API tests (Pytest)** and **Selenium UI tests** for grounded, cited responses, supporting **40+ researchers** at approximately **190ms p95 latency**.
- Built a biomedical ML data pipeline using **Python, SQL, Pandas, AWS S3, and DVC**, curating **30K+ publications** with data validation, deduplication, metadata tagging, and reproducible train/eval splits.
- Fine-tuned **BioBERT** with **PyTorch and Hugging Face** for biomedical NER using **Optuna** and **PEFT/LoRA**, improving **F1 by 18 points** over spaCy across **6K+ annotated sentences**.

Accenture | Machine Learning Engineer **India | September 2020 – July 2023**

- Delivered a **Generative AI** customer support solution with **Azure OpenAI LLMs** and **RAG** for a retail client, handling **15K+ daily queries**; participated in **business requirements reviews** and designed end-to-end test scenarios with product owners.
- Integrated **RAG workflows** with **LangChain**, vector embeddings, and **Pinecone**, validating retrieval and response quality to reduce resolution time **38%** and increase ticket deflection **25%**.
- Wrote **complex Oracle SQL** queries and automated query processing across **Oracle, Databricks, and SageMaker Feature Store** to validate feature data, model metadata, and prediction outputs before each release.
- Built **Selenium and Cucumber (BDD)** automation for web and mobile support workflows, and orchestrated **MLOps** with **Docker, Kubernetes, MLflow, and AWS SageMaker**, accelerating the model deployment lifecycle from **2 weeks to 3 days**.
- Engineered **PySpark and Databricks** pipelines processing **500GB+** of structured and unstructured data, reducing processing time by **45%**, and developed **XGBoost** forecasting models improving accuracy by **22%** across **10+ categories**.
- Established **model monitoring, drift detection**, and automated retraining to maintain **95%+ production accuracy** over **12 months** for **200+ end users**, working in Agile sprints tracked in **Jira** and **Confluence**.

Honeywell | Data Engineer **India | January 2019 – September 2020**

- Engineered **ETL/ELT pipelines** for industrial IoT analytics using **Python, PySpark, and Apache Spark**, ingesting telemetry from **8+ source systems** via REST APIs and processing **2M+ records daily** into **Azure Data Lake**.
- Implemented **data quality and system testing frameworks** (null checks, duplicate detection, schema validation, referential integrity, anomaly detection), reducing downstream reporting errors by **40%** across **3 plant sites**.
- Automated batch workflows with **Azure Data Factory** and **Apache Airflow**, including source-to-target mapping validation and monitoring across **15+ daily jobs**, cutting issue detection time to **under 15 minutes**.
- Built reusable **SQL Server and Oracle** transformation and reconciliation logic with **dbt**, improving data prep efficiency by **35%**, and designed a **star-schema** warehouse that cut report turnaround by **60%** across **5 teams**.
- Managed production deployments with **Git, CI/CD, and Linux/Shell scripting**, archived telemetry in **AWS S3**, and partnered with Power BI developers on **3 on-time dashboard releases**.

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA.

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.